



Directed Works Sheet 3

in Analysis 1

First Year License in Mathematics and Computer Science

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Exercise 1

Using the definition of the limit of a sequence, show that:

$$1. \lim_{n \rightarrow \infty} \frac{3n-1}{2n+3} = \frac{3}{2}$$

$$3. \lim_{n \rightarrow \infty} \frac{2 \ln(1+n)}{\ln n} = 2$$

$$5. \lim_{n \rightarrow \infty} \frac{-5n^2-2}{4n} = -\infty$$

$$2. \lim_{n \rightarrow \infty} \frac{(-1)^n}{2^n} = 0$$

$$4. \lim_{n \rightarrow \infty} 3^n = +\infty$$

$$6. \lim_{n \rightarrow \infty} \ln(\ln n) = +\infty$$

Exercise 2

Calculate the limits of the following sequences given by their general term:

$$1. U_n = \frac{\cos(2n^3-5)}{3n^3+2n^2+1}$$

$$3. U_n = \frac{3n+(-3)^n}{3^n}$$

$$6. U_n = \frac{e^{2n}-e^n+1}{2e^n+3}$$

$$2. U_n = \frac{2n^4-8n^2}{3n^4+\cos n + \frac{1}{n^5}}$$

$$4. U_n = \sqrt{n^2+n+1} - \sqrt{n^2-n+1}$$

$$5. U_n = \frac{3^n - 2.5^n + 6.7^n}{7.2^n + 3.4^n + 5.7^n}$$

$$7. U_n = n^\alpha e^{-\beta n}, \quad \alpha \in \mathbb{R}, \beta \in \mathbb{R}_+^*$$

Exercise 3

- Using the **boundedness principle** for sequences, demonstrate that the sequence $(U_n)_{n \in \mathbb{N}}$ converges and determine its limit l in each case:

$$1. U_n = \sum_{k=0}^n \frac{n}{n^3+k}$$

$$2. U_n = \sum_{k=1}^n \frac{1}{\sqrt{n^2+k}}$$

$$3. U_n = \frac{E(\sqrt{n})}{n}, \text{ where } E(\cdot) \text{ denotes the integer part.}$$

- Let $U_n = \sum_{k=1}^n \frac{1}{3+|\sin k|\sqrt{k}}$, show that $\lim_{n \rightarrow \infty} U_n = +\infty$.

Exercise 4

We consider the sequence $(U_n)_{n \in \mathbb{N}}$ defined by:

$$\begin{cases} U_0 = 0, \\ U_{n+1} = \sqrt{U_n + 2}, \quad \forall n \in \mathbb{N}. \end{cases}$$

1. Show that $0 \leq U_n < 2$, $\forall n \in \mathbb{N}$.
2. Deduce the monotonicity of $(U_n)_{n \in \mathbb{N}}$.

Consider the sequence $(V_n)_{n \in \mathbb{N}}$ defined by:

$$V_n = 2 - U_n, \quad \forall n \in \mathbb{N}.$$

1. What is the sign of V_n ?
2. Show that $\frac{V_{n+1}}{V_n} \leq \frac{1}{2}$, $\forall n \in \mathbb{N}$.
3. Using mathematical induction, show that $V_n \leq \left(\frac{1}{2}\right)^{n-1}$, $\forall n \in \mathbb{N}$.
4. Deduce the limit of the sequence (V_n) , and then the limit of (U_n) .

Exercise 5

Consider the sequence $(U_n)_{n \in \mathbb{N}}$ defined by:

$$\begin{cases} U_0 = 1, \\ U_{n+1} = U_n e^{-U_n}, \quad \forall n \in \mathbb{N}. \end{cases}$$

1. Prove that $U_n > 0$, $\forall n \in \mathbb{N}$.
2. Deduce the monotonicity of the sequence (U_n) .
3. Conclude that (U_n) is convergent, and find its limit.
4. Let $S_n = \sum_{k=0}^n U_k$. Show that $U_{n+1} = e^{-S_n}$, $\forall n \in \mathbb{N}$.
5. Deduce that $\lim_{n \rightarrow \infty} S_n = +\infty$.

Exercise 6

Consider the sequence $(U_n)_{n \in \mathbb{N}}$ defined by:

$$\begin{cases} U_0 > 0, \\ U_{n+1} = \ln(1 + U_n), \quad \forall n \in \mathbb{N}. \end{cases}$$

1. Prove that $0 < U_n$, $\forall n \in \mathbb{N}$.
2. Define $g(x) = \ln(1 + x) - x$; study the variations of g on $]0, +\infty[$ and determine its sign on $]0, +\infty[$.
3. Deduce the monotonicity of (U_n) .
4. Conclude that (U_n) is convergent, and find its limit.
5. Let $E = \{U_n \mid n \in \mathbb{N}\}$; determine $\inf(E)$ and show that $\sup(E)$ is positive.

Exercise 7

Using the **Cauchy criterion**, show that the sequence $(U_n)_{n \in \mathbb{N}^*}$ is convergent and that the sequence $(V_n)_{n \geq 2}$ is divergent:

$$U_n = \sum_{k=0}^n \frac{\sin k}{2^k}, \quad \forall n \in \mathbb{N}^*, \quad V_n = \sum_{k=0}^n \frac{1}{\ln k}, \quad \forall n \geq 2.$$